Overview of Kalman Filter Theory and Navigation Applications

Michael L. Carroll

Feb 27, 2004

©2003 by Michael L. Carroll

Day 5: Details of the Combat Talon I Navigation Kalman Filter

- Segment 1: The Combat Talon I States
- Segment 2: The Combat Talon I Navaids
- Segment 3: The Combat Talon I Kalman Filter High-Level Software Architecture
- Segment 4: The Combat Talon I Kalman Filter JOVIAL Implementation

Day 5, Segment 1
The CT Filter States
Topics

- Historical Overview
- Navigation Modes
- INS Alignment
- The 23 States

Overview of the Combat Talon I Kalman Filter

- Historical Overview
- Structural Overview

Historical Overview

- The Original Ancestor 1970s: F-111
  - Singer-Kearfott (Totowa, NJ)
  - Rockwell Autonetics (Anaheim, CA)
    * Dr. Thomas De Vries

Historical Overview

- Mid-1980s: Combat Talon I SOF Improvements
  - Lockheed Aircraft Services (Ontario, CA)
  - Singer-Kearfott (Totowa, NJ)
    * Dr. George (Bill) Reynaud

Historical Overview

- Siblings and Distant Cousins
  - Fraternal Twin: The Gunship H Kalman Filter
    * First added GPS
    * Lockheed Engineers: Jeff Wyman and Larry Gately

Historical Overview

- Siblings and Distant Cousins
  - Distant Cousin: The C-MIGITS I and II Kalman Filter
    * C/A-code, Miniature Integrated GPS/INS Tactical System
    * descendant of original F-111 filter
    * Rockwell Autonetics (early to mid 1990s)
    * BEI Systron Donner Inertial Division (2000+)
    * C-MIGITS I replace 5-channel Microtraker GPS receiver with 12-channel Jupiter receiver

Historical Overview

- Siblings and Distant Cousins
  - Some Likely Distant Cousins: Various Autonetics filters
    * ICBM
    * Submarine and Ship

Structural Overview

- Two Filters
  - one for each INS
  - two sets of INS reference navigation quantities
- Feed-forward (nav) and feedback mechanizations (alignment)
- Covariance outputs used to construct weights for weighted averaging of navigation quantities

Structural Overview

- Navaids
  - Doppler Velocity Sensor (DVS)
  - Global Positioning System (GPS)
  - True Airspeed Sensor (TAS)

Structural Overview

- Navaids (cont)
  - Baro Altimeter
    - Altitude Calibration
  - Position Fixes
    - Manual
    - Multimode Radar (MMR)

Navigation Modes

- Aided Inertial
  - INS + one or more of the following navaids: GPS/DVS/TAS
  - Position Fixes available in all aided modes
- Unaided (Free) Inertial
- Legacy Mode (no longer available): DVS/TAS - the so-called Doppler filter

Aided Inertial

- Kalman filter operational

- Position Fixes

  - geodetic: filter uses covariance (P) to determine how much posfix delta to allow

  - 100% deterministic

Aided Inertial

- Navigation Kalman filter begins operating when
  - INU is operational and
  - in NAV mode for more than 5 minutes and
  - groundspeed $\leq$ 30 ft/sec (17 kts)
- Alignment filter operates in INU is in either
  - ground fine alignment
  - air alignment

Unaided Inertial

- No Kalman filter running!
- Only 100% posfixes allowed
- Unaided modes
  - Doppler Dead Reckoning (DR)
  - TAS DR
  - GPS
  - No Nav

Alignment

- Stationary Alignment
  - Gyrocompassing
  - Coarse Leveling
  - Fine Alignment
- Motion Alignment
  - Air Alignment
  - Transfer Align

Stationary Alignment

- Motion is bad
  - Don't change the heading!
    - Heading changes are attributed to gyro biases
  - Don't lie to the filter about position!
    - Position drives gyrocompassing and plumb-bob gravity model
  - Allowances made for wind buffeting in process noise
- Gyrocompassing: find north through sensing of earth rate

Motion Alignment

- Typically In-Air
- Coarse and Fine Alignment Phases
- Velocity Aided
  - GPS (but position as well!)
  - DVS
  - Other INS (transfer align)

The 23 States of the Combat Talon I Kalman Filter

- 11 Time Varying States
- 12 Non-Time Varying States

Combat Talon I Kalman Filter States: 11 Time Varying States

- 9 Canonical Inertial Error States
- 2 Baro Inertial Loop States

11 Time-Varying States (P$KALTYPES.CPL)

```
308  TYPE INS'SV'VAR'ITEM   TABLE;
309    BEGIN
310       ITEM  KAL'INS'X'POS'ERROR                    REAL;
311       ITEM  KAL'INS'Y'POS'ERROR                    REAL;
312       ITEM  KAL'INS'Z'POS'ERROR                    REAL;
313       ITEM  KAL'INS'X'VEL'ERROR                    REAL;
314       ITEM  KAL'INS'Y'VEL'ERROR                    REAL;
315       ITEM  KAL'INS'Z'VEL'ERROR                    REAL;
316       ITEM  KAL'INS'X'TILT'ERROR                   REAL;
317       ITEM  KAL'INS'Y'TILT'ERROR                   REAL;
318       ITEM  KAL'INS'HEADING'ERROR                  REAL;
319       ITEM  KAL'INS'BARO'ALTI'ERROR                REAL;
320       ITEM  KAL'INS'VERT'CHANNEL'ERROR             REAL;
321    END
```

Combat Talon I Kalman Filter States: Canonical 9-State Navigation Errors (SRS Symbology)

- Position Errors: $\Delta RX, \Delta RY, \Delta RZ$
- Velocity Errors: $\epsilon VX, \epsilon VY, \epsilon VZ$
- Attitude Errors: $\Theta X, \Theta Y, \Psi Z$

Combat Talon I Kalman Filter States: Baro-Inertial Loop States

- Baro Altimeter Bias: $\epsilon\Delta H$
- 3rd-order baro-inertial loop error state: $I\Delta H$

Combat Talon I Kalman Filter States: 12 Non-Varying States

- 3 Gyro Errors
- 6 Doppler Errors
- 3 Accelerometer Errors

12 Non-Varying States (P$KALTYPES.CPL)

```
325   TYPE INS'SV'FIX'ITEM     TABLE;
326    BEGIN
327       ITEM  KAL'INS'X'GYRO'BIAS'ERROR          REAL;
328       ITEM  KAL'INS'Y'GYRO'BIAS'ERROR          REAL;
329       ITEM  KAL'INS'Z'GYRO'BIAS'ERROR          REAL;
330       ITEM  KAL'INS'DOP'X'SF'ERROR             REAL;
331       ITEM  KAL'INS'DOP'Z'BS'ERROR             REAL;
332       ITEM  KAL'INS'DOP'Y'BS'ERROR             REAL;
333       ITEM  KAL'INS'DOP'X'B'ERROR              REAL;
334       ITEM  KAL'INS'DOP'Y'B'ERROR              REAL;
335       ITEM  KAL'INS'DOP'Z'B'ERROR              REAL;
336       ITEM  KAL'INS'X'ACC'BIAS'ERROR           REAL;
337       ITEM  KAL'INS'Y'ACC'BIAS'ERROR           REAL;
338       ITEM  KAL'INS'Z'ACC'BIAS'ERROR           REAL;
339    END
```

Combat Talon I Kalman Filter States: 3 INS Gyro Bias Errors

- $\epsilon DX, \epsilon DY, \epsilon DZ$
- Modeled as random constants: $\dot{x} = 0$
- Used in computing tilt derivatives in 4 Hz slow nav extrapolation: SN'INS'ERROR'EXTRAP

Combat Talon I Kalman Filter States: 6 Doppler Errors

- X scale factor error: $\epsilon SFDX$
- Z and Y boresight errors: $\delta BZ, \delta BY$
- X, Y, Z bias errors: $\epsilon BDX, \epsilon BDY, \epsilon BDZ$
- Used in compensating Doppler velocity observations

Combat Talon I Kalman Filter States: 3 INS Accelerometer Bias Errors

- $\epsilon BX, \epsilon BY, \epsilon BZ$

- Used in computing velocity derivatives in 4 Hz slow nav extrapolation: SN'INS'ERROR'EXTRAP

Day 5, Segment 2
Combat Talon Navaids
Topics

- The CT Sensors and their Intefaces

- Sensor Processing

The Sensors and their Interfaces

- Global Positioning System (GPS)
- Doppler Velocity Sensor (DVS)
- Air Data Computer
- Multimode Radar (MMR)

GPS

- AN/APN-151
- Rockwell-Collins 3A Receiver
- 5 states
- P-code

GPS Interfaces

- 1553B
- GPS receives: I1, I2, I4, I5A, I6, I15, I22
- GPS transmits: G01, G02, G03, G09, G13

Doppler Velocity Sensor (DVS)

- AN/APN-218
- Originally from Canadian Marconi

DVS Interfaces

- Direct Device I/O
- ARINC 575 serial input
- heading, drift and vertical velocity
- velocity validites
- sensor validity message

Air Data Computer (ADC)

- LMSW P/N 4065084-3
- Integrated externally with pitot-static and temperature sub-systems

ADC Interfaces

- Airspeed
- Baro Alt (coarse and fine)
- Validity signals

Multimode Radar (MMR)

- AN-APQ-122(V)8
- Texas Instruments (now Raytheon)
- Provides standalone terrain following functions in addition to mapping

MMR Interfaces

- MMR outputs: X-Y Joystick serial data
- MMR inputs: drift angle, azimuth stability, validity signals

Navaid Measurement Processing

- Differencing against inertial reference quantities
- Detailed discussion deferred until code discussion